CS6460 Ed Tech: Project Proposal

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1 INTRODUCTION

Mandarin Chinese is the second most widely spoken language in the world, yet it presents one of the most persistent challenges for non-native learners: the acquisition of lexical tones. Unlike Indo-European languages, Mandarin encodes word meaning through pitch contours, with four distinct tones that can completely alter the semantic content of an otherwise identical syllable. For learners whose native language does not use pitch lexically - including speakers of English and most South Asian languages - this tonal dimension represents a fundamental discontinuity from their prior linguistic experience. Even after extended study, many learners plateau in tone accuracy, particularly in continuous speech and naturally occurring conversational contexts.

This challenge is not simply a matter of exposure or effort. Research in second language acquisition (SLA) has established that tone perception and production involve distinct cognitive mechanisms, and that learners from non-tonal language backgrounds must develop entirely new phonological representations rather than mapping Mandarin tones onto existing L1 categories (Wang, Jongman, & Sereno, n.d.; Zou, 2017). Theoretical frameworks including the Perceptual Assimilation Model (Tyler & Best, 2024), the Speech Learning Model (Hao, 2014), and the L2 Linguistic Perception Model (van Leussen & Escudero, 2015) each offer partial accounts of why tone acquisition is difficult, but together they reinforce a shared conclusion: learners require structured, perceptually grounded feedback in order to notice and correct tonal errors - feedback that classroom instruction alone rarely provides at sufficient scale or frequency.

Technology-assisted pronunciation training has long been proposed as a solution to this instructional gap. Computer-assisted pronunciation training (CAPT) systems can offer self-paced, repeatable practice with immediate feedback, addressing the shortage of individualized instruction that classroom settings impose (Amrate & Tsai, 2025). Visual feedback tools such as Praat-based pitch contour displays have demonstrated measurable improvements in learner tone production under controlled conditions (Chen, 2024; Chun, Jiang, & Ávila, 2013),

and more recent work on deep learning-based mispronunciation detection - including BLSTM-based models with soft-target tone labels (Li et al., 2019) and Siamese network architectures for pitch contour comparison (Bu et al., 2025) - suggests that automatic tone assessment is technically feasible at a level of accuracy suitable for learner-facing applications.

However, a critical gap persists between what these technologies can detect and what helps learners improve. Existing tools disproportionately focus on segmental accuracy evaluated against native-speaker norms, while neglecting suprasegmental features such as tone contour shape, coarticulation, and contextual tone variation that are central to intelligibility (Rogerson-Revell, 2021). Feedback is frequently represented in technical formats - spectrograms, F0 curves, confidence scores - that learners cannot interpret without guidance, reducing its pedagogical value even when technically accurate (Amrate & Tsai, 2025). Furthermore, studies on learner experience consistently show that perceived usefulness, interpretability of feedback, and usability are as important as technical accuracy in determining whether learners adopt and persist with pronunciation tools (Davis, 1989; Yuen & Schlote, 2024).

This project addresses that gap by developing a learner-centered, web-based tool for Mandarin tone perception and production training. The tool provides non-native learners with real-time tone classification feedback, visual pitch contour comparison between their productions and native speaker references, and interpretable, actionable guidance designed around established principles from both SLA theory and user experience design. Unlike existing research prototypes, which are largely confined to laboratory settings or require instructor mediation, this tool is designed for autonomous, out-of-class practice by learners with no prior phonetic training. By grounding both the feedback design and the evaluation methodology in empirically supported principles, this project aims to move beyond technical accuracy as the sole criterion of success and toward a tool that demonstrably supports learner improvement.

2 RELATED WORKS

Research on technology-assisted Mandarin tone learning spans three broad areas relevant to this project: computer-assisted pronunciation training (CAPT) systems, automatic tone detection and mispronunciation identification, and

multimodal feedback design. Together, these bodies of work establish both the technical feasibility of automated tone feedback and the persistent gap between system capability and learner-centered design.

2.1 Computer-Assisted Pronunciation Training for Mandarin Tones

CAPT systems have been used to support Mandarin tone learning for over two decades, with visual pitch contour feedback emerging as one of the most consistently effective approaches. Qin and Wang (2014) developed an early computer-aided Chinese pronunciation training system that extracted pitch contours from learner speech and displayed them alongside native speaker references, demonstrating measurable improvements in tone production among L2 learners. Chen (2024) extended this line of work using Praat, showing that feedback incorporating learners' own recorded voices produced greater improvement than feedback based on a model speaker's voice alone, and that combining audio and visual feedback outperformed either modality in isolation. Chun, Jiang, and Ávila (2013) similarly reported approximately 50% improvement in tonal pronunciation among learners who used side-by-side pitch contour comparisons, with participants reporting that visual feedback helped them perceive and correct errors they could not detect by ear alone.

Despite these promising results, existing Praat-based tools share a common limitation: they require instructor mediation, offline analysis, and controlled recording conditions, making them poorly suited for autonomous, out-of-class learner practice. A systematic review of CAPT research by Amrate and Tsai (2025) found that most systems continue to evaluate learner speech against native-speaker norms, with limited attention to suprasegmental features and little accommodation of diverse learner backgrounds. Rogerson-Revell (2021) further notes that feedback representations such as spectrograms and F0 curves frequently exceed learners' interpretive capacity without scaffolding, limiting their pedagogical value even when technically accurate.

2.2 Automatic Tone Detection and Mispronunciation Identification

On the technical side, significant advances have been made in automatic tone classification and mispronunciation detection for Mandarin. Li et al. (2019) demonstrated that BLSTM-based deep tone models using soft-target tone labels substantially outperform traditional hard-label classifiers in detecting tonal

errors, particularly for tones with high confusion rates. Bu et al. (2025) introduced a Siamese neural network architecture that compares learner pitch contours against native references using a learned similarity metric, achieving strong robustness to speaker variation and recording conditions. More recently, Gan and Wei (2025) proposed a BIDFSMN-based CAPT system achieving approximately 90% detection accuracy on Mandarin pronunciation errors in non-native speakers.

A recurring finding across these studies is that modeling tonal context - the influence of neighboring tones on each other through coarticulation - substantially improves detection accuracy over systems that classify tones in isolation (Tang & Li, 2021; Tong et al., 2018). However, these technical systems are evaluated almost exclusively on classification metrics such as accuracy and F1 score, with little attention to how detected errors are communicated to learners or whether the feedback supports measurable improvement in production.

2.3 Multimodal Feedback and Learner Experience

A separate strand of research has examined how feedback modality affects tone acquisition outcomes. Godfroid, Lin, and Ryu (2017) demonstrated that color-coded visual representations of lexical tones, when combined with auditory input, produced significantly greater perceptual gains than audio-only training, particularly for difficult tone contrasts. Farran and Morett (2024) review evidence that beginner learners benefit from visual pitch representations before relying solely on auditory cues, suggesting a developmental role for multimodal feedback in early tone acquisition. Meng et al. (2023) showed that perception-guided learning - where the system optimises for the learner's ability to perceive their own errors rather than acoustic matching to a native target - produces stronger tone learning outcomes than contour-matching feedback alone.

Learner experience research highlights that technical capability alone does not determine adoption or persistence. Studies of language learning app usage consistently find a gap between perceived usefulness and sustained engagement, with usability, feedback interpretability, and learner trust emerging as critical mediating factors (Yuen & Schlote, 2024; Leva et al., 2025). Liu et al. (2025) specifically found that time pressure during voice input tasks significantly

increases cognitive load and error rates for Mandarin learners, reinforcing the importance of learner-controlled pacing in pronunciation tool design.

3 PROPOSED WORKS

This project develops a web-based, learner-centered tool for Mandarin tone perception and production training, targeting non-native speakers of Mandarin Chinese. The tool integrates automatic tone classification with visual pitch contour feedback, designed for autonomous out-of-class practice without instructor mediation. The following subsections describe the system design, technical implementation, and evaluation methodology.

3.1 System Overview

The tool presents learners with a Mandarin word, prompts them to record their pronunciation, classifies the produced tone, and displays a visual comparison of the learner's pitch contour against a native speaker reference. Feedback is delivered in plain language alongside the visual display, identifying which tone was detected, whether it matches the target, and what the learner should adjust. The interaction is self-paced, with no time pressure, consistent with findings by Liu et al. (2025) showing that constrained response timing significantly increases cognitive load and error rates for Mandarin learners.

The tool targets the four canonical Mandarin tones (T1–T4) in monosyllabic and disyllabic word contexts. Disyllabic words are included specifically to expose learners to tone sandhi - the systematic modification of Tone 3 before another Tone 3, which is one of the most common sources of error for non-native learners and is absent from isolated syllable practice. Continuous speech is outside the scope of this project and is identified as a direction for future work.

3.2 Technical Implementation

Tech Stack:

The tool will be built using Python with a Streamlit frontend. Streamlit provides built-in audio recording support, rapid prototyping capability, and easy public deployment via Streamlit Community Cloud, making it well suited for a course project that requires a shareable, demonstrable artifact. All processing - audio

input, pitch extraction, tone classification, and feedback generation will be handled in the Python backend.

Pitch Extraction and Visualization:

Pitch contour extraction will be performed using Parselmouth, a Python library that wraps the Praat speech analysis engine. Parselmouth extracts fundamental frequency (F0) values over time from a recorded audio signal, producing the pitch contour that forms the basis of both the visual feedback and the classification features. This approach is directly grounded in the Praat-based methods demonstrated effective in the literature (Chen, 2024; Chun, Jiang, & Ávila, 2013; Meng et al., 2023).

Following recording, the learner's pitch contour will be displayed alongside a pre-extracted native speaker reference contour for the same word, rendered as an overlay plot using Matplotlib or Plotly. The two contours will be normalised to a common time axis and semitone scale to account for differences in speaking rate and pitch range between speakers - a standard technique in phonetic analysis that makes cross-speaker comparison interpretable. Feedback will be displayed after recording is complete rather than in real time, consistent with the post-recording design validated by Chen (2024) and appropriate for the technical constraints of a browser-based implementation.

Tone Classification:

Tone classification will be implemented using acoustic features extracted via Parselmouth specifically, F0 contour shape features including mean F0, F0 slope, F0 range, and contour curvature fed into a lightweight classifier. A Support Vector Machine (SVM) or small feedforward neural network will be trained on tone-labelled Mandarin speech to classify the learner's produced tone as one of T1–T4. This approach is computationally lightweight, does not require GPU inference, and produces interpretable feature representations that can be described clearly in the final paper. It is consistent with the Praat-based classification tradition in CAPT research and avoids the overfitting risk of fine-tuning a large model like wav2vec2 on a limited labelled dataset within the project timeframe.

Dataset:

Native speaker reference audio and classifier training data will be sourced from the AISHELL-1 corpus, a freely available, large-scale Mandarin speech dataset with multiple speakers and high audio quality, widely used in the literature (Tang & Li, 2021). Tone labels will be derived from the pinyin annotations included in the AISHELL metadata. For the tone classification component, a balanced subset of AISHELL-1 will be used, sampling across speakers and tonal contexts to improve robustness. The Bu et al. (2025) corpus, which contains both native and non-native Mandarin speech with tone accuracy annotations, will be used for evaluation of the classifier on non-native speech if access can be confirmed; otherwise, a small supplementary dataset of self-recorded non-native speech will be used for this purpose.

Feedback Design:

The feedback interface will display three elements: the overlay pitch contour plot, a plain-language tone identification statement (e.g., "Your pitch contour suggests Tone 2 — the target is Tone 4"), and a brief, actionable correction cue describing the shape difference (e.g., "Tone 4 should fall sharply from high to low - your contour stayed level"). This design is informed by the distinction between diagnostic transparency and learner interpretability identified across the literature: raw acoustic displays without explanation have been shown to confuse rather than support learners (Rogerson-Revell, 2021; Amrate & Tsai, 2025), while explicit, plain-language guidance scaffolds noticing in a way consistent with Schmidt's (1990) Noticing Hypothesis.

3.3 Word List and Content Design

The tool will include a curated word list of approximately 60–80 Mandarin words, covering all four tones in both monosyllabic and disyllabic contexts and sampled to include common vocabulary relevant to beginner and intermediate learners. Disyllabic pairs will include tone sandhi contexts (T3+T3 combinations) to address a well-documented source of learner error. Words will be presented in pinyin alongside Chinese characters, with an English gloss, to avoid conflating character recognition difficulty with tone production difficulty.

3.4 Evaluation

The tool will be evaluated along two dimensions: technical performance of the tone classifier and learner usability and experience.

Technical Evaluation:

The tone classifier will be evaluated on a held-out test set using accuracy, per-class F1 score, and a confusion matrix across T1–T4. The confusion matrix is particularly informative for this task, as certain tone pairs (T2/T3 in particular) are known to be systematically more confusable for both learners and classifiers. Results will be reported separately for native and non-native speaker test speech where data permits.

Learner Evaluation:

A small usability study will be conducted with approximately 6–10 non-native Mandarin learners recruited from the Georgia Tech student community or online language learning communities. Participants will complete a short pre-session tone perception test (identifying the tone of 20 audio clips), use the tool for a single 20–30 minute practice session, and complete a post-session tone perception test using a matched set of 20 clips. They will also complete a short post-session survey measuring perceived usefulness, ease of use, and feedback interpretability, adapted from the Technology Acceptance Model items established by Davis (1989). Pre/post perception scores will be compared to assess whether a single session of tool use produces measurable gains in tone identification accuracy. Qualitative feedback from the survey will be used to identify usability issues and inform design recommendations.

This evaluation design is intentionally modest in scope and appropriate for a 10-week solo project. It does not claim to measure long-term learning outcomes or production gains, which would require a longitudinal design outside the project's timeframe. These are identified as directions for future work.

3.5 Limitations and Fallback Plans

Several risks are acknowledged. If the Bu et al. (2025) corpus is not publicly accessible, evaluation on non-native speech will rely on a smaller self-recorded dataset, which will be noted as a limitation in the final paper. If Streamlit's audio recording functionality proves unreliable in the target browser environment, the

interface will be reimplemented using Gradio, which offers comparable functionality with a different underlying audio handling approach. If participant recruitment for the user study falls below six participants, the evaluation will shift to a purely technical assessment supplemented by a self-evaluation of the tool against the usability heuristics established in the literature.

4 DELIVERABLES

This project will produce a series of deliverables across the semester, culminating in a functional web-based Mandarin tone training tool, a research paper, and a final presentation. The key milestones are described below.

Intermediate Milestone 1 (Week 10, due March 22, 2026)

By the first intermediate milestone, the core technical pipeline will be implemented and demonstrable. This includes a working Streamlit interface that accepts learner audio input, extracts pitch contours using Parselmouth, and displays a visual overlay of the learner's contour against a native speaker reference from the AISHELL-1 corpus. The tone classifier will be trained on a subset of AISHELL-1 and capable of classifying T1–T4 for monosyllabic words, with preliminary accuracy results reported. The word list will be finalized and integrated into the interface. At this stage, the tool will be functional end-to-end, though the feedback language, UI polish, and evaluation study will not yet be complete. The milestone video will demonstrate a live recording session, the pitch contour visualization output, and the tone classification result, and will specifically solicit feedback on the clarity of the visual feedback display and the interpretability of the plain-language correction cues.

Intermediate Milestone 2 (Week 13, due April 12, 2026)

By the second intermediate milestone, the tool will be complete and the learner evaluation study will be underway or concluded. This includes the full word list covering monosyllabic and disyllabic words with tone sandhi contexts, the finalized feedback interface with plain-language correction guidance, and completed pre/post tone perception tests and TAM-based usability surveys from participants. Technical evaluation results - accuracy, per-class F1, and confusion matrix across T1–T4 will be available. The milestone video will present the complete tool, summarize preliminary evaluation findings, and solicit feedback

on the evaluation methodology and any observed usability issues that remain to be addressed before final submission.

Final Project Deliverables (Week 16, due May 3, 2026)

The final submission will consist of four components:

The Tool:

A fully functional, publicly deployed Streamlit web application for Mandarin tone perception and production training, including the complete word list, pitch contour visualisation, tone classifier, and plain-language feedback system. The application will be accessible via a public Streamlit Community Cloud URL and accompanied by a GitHub repository containing all source code, the curated word list, pre-extracted native speaker reference contours, and instructions for local deployment.

The Final Paper:

A written report in JDF format documenting the motivation and problem statement, related work, system design and implementation decisions, evaluation methodology and results, limitations, and directions for future work. The paper will report both technical classifier performance and learner evaluation findings, including pre/post tone perception scores and TAM survey results.

The Final Presentation:

A recorded or live presentation summarizing the project, demonstrating the tool, and discussing key findings and lessons learned.

Weekly Status Reports:

Eight weekly status check submissions (Weeks 8–15) documenting progress, challenges encountered, and any revisions to the project plan.

5 TASK LIST

Week #	Task #	Task Description	Estimated Time (Hours)	Member Responsible			
8	1	Set up Python/Streamlit development environment	2	Bharath			
8	2	Download and explore AISHELL-1 corpus	2	Bharath		**Total Hours**	**101.5**
8	3	Write scripts to extract tone-labelled audio segments from AISHELL-1	3	Bharath			
8	4	Verify access to Bu et al. (2025) corpus; pursue alternative if unavailable	1	Bharath		Bharath	101.5
8	5	Pre-extract native speaker reference pitch contours using Parselmouth	2	Bharath			
9	1	Extract F0-based acoustic features from AISHELL-1 training set	3	Bharath			
9	2	Train and evaluate SVM/feedforward classifier on T1-T4	4	Bharath			
9	3	Generate confusion matrix and per-class F1 scores	2	Bharath			
9	4	Iterate on feature engineering to improve classification accuracy	2	Bharath			
9	5	Write Weekly Status Check 2	2	Bharath			
10	1	Build Streamlit audio recording interface	3	Bharath			
10	2	Integrate Parselmouth pitch extraction into recording pipeline	2	Bharath			
10	3	Implement pitch contour overlay visualisation (learner vs. native reference)	3	Bharath			
10	4	Connect tone classifier to interface for end-to-end demo	2	Bharath			
10	5	Finalise monosyllabic word list	1.5	Bharath			
10	6	Prepare and submit Intermediate Milestone 1 video	2	Bharath			
10	7	Write Weekly Status Check 3	2	Bharath			
INTERMEDIATE MILESTONE 1 DUE							
11	1	Write plain-language correction cues for all T1-T4 error combinations	3	Bharath			
11	2	Extend word list to disyllabic words including tone sandhi contexts	2	Bharath			
11	3	Integrate disyllabic words into interface	2	Bharath			
11	4	Refine UI layout and feedback display based on Milestone 1 feedback	2	Bharath			
11	5	Write Weekly Status Check 4	2	Bharath			
12	1	Design pre/post tone perception test (20 audio clips each)	3	Bharath			
12	2	Develop TAM-based post-session usability survey	2	Bharath			
12	3	Prepare participant instructions and consent materials	2	Bharath			
12	4	Begin participant recruitment	2	Bharath			
12	5	Write Weekly Status Check 5	2	Bharath			
13	1	Run evaluation sessions with 6-10 participants	4	Bharath			
13	2	Collect pre/post perception test scores and survey responses	2	Bharath			
13	3	Begin preliminary analysis of evaluation data	3	Bharath			
13	4	Prepare and submit Intermediate Milestone 2 video	2	Bharath			
13	5	Write Weekly Status Check 6	2	Bharath			
INTERMEDIATE MILESTONE 2 DUE							
14	1	Complete analysis of technical and learner evaluation results	4	Bharath			
14	2	Draft paper: Introduction, Related Work, Proposed Work sections	5	Bharath			
14	3	Write Weekly Status Check 7	2	Bharath			
15	1	Draft and finalise paper: Results, Discussion, Limitations, Future Work	5	Bharath			
15	2	Revise and polish full paper draft	3	Bharath			
15	3	Prepare final presentation slides or recording	2	Bharath			
15	4	Write Weekly Status Check 8	2	Bharath			
16	1	Final review and polish of tool, paper, and presentation	2	Bharath			
16	2	Deploy tool to Streamlit Community Cloud	1	Bharath			
16	3	Ensure GitHub repository is clean and documented	1	Bharath			
16	4	Submit project, paper, and presentation	1	Bharath			
FINAL PROJECT DUE							

6 REFERENCES

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